LAMOGA-25
LAMOGA-50
LAMOGA-100

ANTIEPILEPTIC

Each film coated tablet contains:
Lamotrigine 25 mg
Lamotrigine 50 mg
Lamotrigine 100 mg

✧ PRODUCT DESCRIPTION:

25 mg:
Pale yellow, arc-square film coated tablet with plain flat face on one side and slope edge, flat face with engraved 25 on the other side

50 mg:
Pale yellow, arc-square film coated tablet with plain flat face on one side and slope edge, flat face with engraved 50 on the other side

100 mg:
Pale yellow, arc-square film coated tablet with plain flat face on one side and slope edge flat face with engraved 100 on the other side

✧ MECHANISM OF ACTION:

***Pharmacology*[3]**
It is suggested that Lamotrigine is a use- and voltage-dependent blocker of voltage gated sodium channels. It inhibits sustained repetitive firing of neurons and inhibits release of glutamate (the neurotransmitter which plays a key role in the generation of epileptic seizures). These effects are likely to contribute to the anticonvulsant properties of Lamotrigine.
In contrast, the mechanisms by which Lamotrigine exerts its therapeutic action in bipolar disorder have not been established, although interaction with voltage gated sodium channels is likely to be important.

Pharmacokinetics
Absorption
Lamotrigine is rapidly and completely absorbed from the gut with no significant first-pass metabolism. Peak plasma concentrations occur approximately 2.5 hours after oral administration of Lamotrigine. Time to maximum concentration is slightly delayed after food but the extent of absorption is unaffected. There is considerable inter-individual variation in steady state maximum concentrations but within an individual, concentrations rarely vary.
Distribution
Binding to plasma proteins is about 55%; it is very unlikely that displacement from plasma proteins would result in toxicity.
The volume of distribution is 0.92 to 1.22 L/Kg.
Biotransformation
UDP-glucuronyl transferases have been identified as the enzymes responsible for metabolism of Lamotrigine. Lamotrigine induces its own metabolism to a modest extent depending on dose. However, there is no evidence that Lamotrigine affects the pharmacokinetics of other AEDs and data suggest that interactions between Lamotrigine and medicinal products metabolized by cytochrome P450 enzymes are unlikely to occur.
Elimination
The apparent plasma clearance in healthy subjects is approximately 30 mL/min. Clearance of Lamotrigine is primarily metabolic with subsequent elimination of glucuronide-conjugated material in urine. Less than 10% is excreted unchanged in the urine. Only about 2% of Lamotrigine-related material is excreted in feces. Clearance and half-life are independent of dose. The apparent plasma half-life in healthy subjects is estimated to be approximately 33 hours (range 14 to 103 hours).
The half-life of Lamotrigine is greatly affected by concomitant medicinal products. Mean half-life is reduced to approximately 14 hours when given with glucuronidation-inducing medicinal products such as Carbamazepine and Phenytoin and is increased to a mean of approximately 70 hours when co-administered with Valproate alone (see Dosage and administration).
Linearity
The pharmacokinetics of Lamotrigine are linear up to 450 mg, the highest single dose tested.
Special patient populations
Children
Clearance adjusted for body weight is higher in children than in adults with the highest values in children under five years. The half-life of Lamotrigine is generally shorter in children than in adults with a mean value of approximately 7 hours when given with enzyme-inducing medicinal products such as Carbamazepine and Phenytoin and increasing to mean values of 45 to 50 hours when co-administered with Valproate alone (see Dosage and Administration).
Infants aged 2 to 26 months
The predicted serum concentration levels in children of 2 to 26 months were in general in the same range as those in older children, though higher Cmax levels are likely to be observed in some children with a body weight below 10 Kg.
Elderly
Results of a population pharmacokinetic analysis including both young and elderly patients with epilepsy, enrolled in the same trials, indicated that the clearance of Lamotrigine did not change to a clinically relevant extent. After single doses apparent clearance decreased by 12% from 35 mL/min at age 20 to 31 mL/min at 70 years. The decrease after 48 weeks of treatment was 10% from 41 to 37 mL/min between the young and elderly groups. In addition, pharmacokinetics of Lamotrigine was studied in 12 healthy elderly subjects following a 150 mg single dose. The mean clearance in the elderly (0.39 mL/min/Kg) lies within the range of the mean clearance values (0.31 to 0.65 mL/min/Kg) obtained in nine studies with non-elderly adults after single doses of 30 to 450 mg.
Renal impairment
For this patient population, initial doses of Lamotrigine should be based on the patient's concomitant medicinal products; reduced maintenance doses may be effective for patients with significant renal functional impairment (see Dosage and Administration and Precaution).
Hepatic impairment
Initial, escalation and maintenance doses should generally be reduced in patients with moderate or severe hepatic impairment (see Dosage and Administration).

✧ INDICATION[1]:

Epilepsy
Adults
Lamotrigine is indicated for use as adjunctive or monotherapy in the treatment of epilepsy, for partial seizures and generalized seizures, including tonic-clonic seizures and the seizures associated with Lennox-Gastaut Syndrome.
Children (2 to 12 years of age)
Lamotrigine is indicated as adjunctive therapy in the treatment of epilepsy, for partial seizures and generalized seizures including tonic-clonic seizures and the seizures associated with Lennox-Gastaut syndrome (above 3 years of age only).
Initial monotherapy treatment in newly diagnosed pediatric patients is not recommended.
After epileptic control has been achieved during adjunctive therapy, concomitant anti-epileptic drugs (AEDs) may be withdrawn and patients continued on Lamotrigine monotherapy.
Bipolar disorder
Adults (18 years of age and over)
Lamotrigine is indicated for the prevention of mood episodes in patients with bipolar disorder, predominantly by preventing depressive episodes.

✧ DOSAGE AND ADMINISTRATION[1]:

Lamotrigine tablets should be swallowed whole, and should not be chewed or crushed.
If a calculated dose of Lamotrigine e.g. for use in children (epilepsy only) or patients with hepatic impairment, cannot be divided into multiple lower strength tablets, the dose to be administered is that equal to the nearest lower strength of whole tablets.
Restarting Therapy
Prescribers should assess the need for escalation to maintenance dose when restarting Lamotrigine in patients who have discontinued Lamotrigine for any reason; since the risk of serious rash is associated with high initial doses and exceeding the recommended dose escalation for Lamotrigine (see Warning and precaution). The greater the interval of time since the previous dose, the more consideration should be given to escalation to the maintenance dose. When the interval since discontinuing Lamotrigine exceeds five half-lives (see Pharmacokinetics), Lamotrigine should generally be escalated to the maintenance dose according to the appropriate schedule.
It is recommended that Lamotrigine not be restarted in patients who have discontinued due to rash associated with prior treatment with Lamotrigine unless the potential benefit clearly outweighs the risk.
Epilepsy
When concomitant anti-epileptic drugs are withdrawn to achieve Lamotrigine monotherapy or other AEDs are added-on to treatment regimes containing Lamotrigine, consideration should be given to the effect this may have on Lamotrigine pharmacokinetics (see Drug interaction).
Dosage in Epilepsy Monotherapy
Adults (over 16 years of age) (see Table 1)
The initial Lamotrigine dose in monotherapy is 25 mg once a day for two weeks, followed by 50 mg once a day for two weeks. Thereafter, the dose should be increased by a maximum of 50 to 100 mg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 100 to 200 mg/day given once a day or as two divided doses. Some patients have required 500 mg/day of Lamotrigine to achieve the desired response.
Because of a risk of rash the initial dose and subsequent dose escalation should not be exceeded (see Warning and precaution).
Dosage in Epilepsy Add-On Therapy
Adults (over 12 years of age) (see Table 1)
In patients taking Valproate with/without any other AED, the initial Lamotrigine dose is 25 mg every alternate day for two weeks, followed by 25 mg once a day for two weeks.
Thereafter, the dose should be increased by a maximum of 25 to 50 mg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 100 to 200 mg/day given once a day or in two divided doses.
In those patients taking concomitant AEDs or other medications (see Drug interaction) that induce Lamotrigine glucuronidation with/without other AEDs (except Valproate), the initial Lamotrigine dose is 50 mg once a day for two weeks, followed by 100 mg/day given in two divided doses for two weeks.
Thereafter, the dose should be increased by a maximum of 100 mg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 200 to 400 mg/day given in two divided doses.
Some patients have required 700 mg/day of Lamotrigine to achieve the desired response.
In those patients taking other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction), the initial Lamotrigine dose is 25 mg once a day for two weeks, followed by 50 mg once a day for two weeks. Thereafter, the dose should be increased by a maximum of 50 to 100 mg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve an optimal response is 100 to 200 mg/day given once a day or as two divided doses.

Table 1: Recommended Treatment Regimen in Epilepsy for Adults

Treatment regimen		Weeks 1 - 2	Weeks 3 - 4	Maintenance Dose
Monotherapy (over 16 years of age)		25 mg (once a day)	50 mg (once a day)	100 - 200 mg (once a day or two divided doses) To achieve maintenance, doses may be increased by 50 - 100 mg every one to two weeks.
Add-on therapy with Valproate regardless of any concomitant medications (over 12 years of age)		12.5 mg (given 25 mg alternate days)	25 mg (once a day)	100-200 mg (once a day or two divided doses) To achieve maintenance, doses may be increased by 25-50 mg every one to two weeks.
Add-on therapy without Valproate	This dosage regimen should be used with: Phenytoin Carbamazepine Phenobarbitone Primidone Or with other inducers of Lamotrigine glucuronidation (see Drug interaction)	50 mg (once a day)	100 mg (two divided doses)	200-400 mg (two divided doses) To achieve maintenance, doses may be increased by 100 mg every one to two weeks.
	This dosage regimen should be used with other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction)	25 mg (once a day)	50 mg (once a day)	100-200 mg (once a day or two divided doses) To achieve maintenance, doses may be increased by 50 - 100 mg every one to two weeks.
In patients taking AEDs where the pharmacokinetic interaction with Lamotrigine is currently not known (see Drug interaction), the treatment regimen as recommended for Lamotrigine with concurrent Valproate should be used.				

Because of the risk of rash the initial dose and subsequent dose escalation should not be exceeded (See Warning and precaution).
Children (2 to 12 years of age) (see Table 2)
In patients taking Valproate with/without any other AED, the initial Lamotrigine dose is 0.15 mg/Kg bodyweight/day given once a day for two weeks, followed by 0.3 mg/Kg/day once a day for two weeks. Thereafter, the dose should be increased by a maximum of 0.3 mg/Kg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 1 to 5 mg/Kg/day given once a day or in two divided doses, with a maximum of 200 mg/day.
In those patients taking concomitant AEDs or other medications (see Drug interaction) that induce Lamotrigine glucuronidation with/without other AEDs (except Valproate), the initial Lamotrigine dose is 0.6 mg/Kg bodyweight/day given in two divided doses for two weeks, followed by 1.2 mg/Kg/day given in two divided doses for two weeks. Thereafter, the dose should be increased by a maximum of 1.2 mg/Kg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 5 to 15 mg/Kg/day given in two divided doses, with a maximum of 400 mg/day.
In patients taking other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction), the initial Lamotrigine dose is 0.3 mg/Kg bodyweight/day given once a day or in two divided doses for two weeks, followed by 0.6 mg/Kg/day given once a day or in two divided doses for two weeks. Thereafter, the dose should be increased by a maximum of 0.6 mg/Kg every one to two weeks until the optimal response is achieved. The usual maintenance dose to achieve optimal response is 1 to 10 mg/Kg/day given once a day or in two divided doses, with a maximum of 200 mg/day.
To ensure a therapeutic dose is maintained the weight of a child must be monitored and the dose reviewed as weight changes occur.

Table 2: Recommended Treatment Regimen in Epilepsy for Children Aged 2-12 years (Total Daily Dose in mg/Kg Body weight/ day)

Treatment regimen		Weeks 1 - 2	Weeks 3 - 4	Maintenance Dose
Add-on therapy with Valproate regardless of any concomitant medications		0.15 mg/Kg* (once a day)	0.3 mg/Kg (once a day)	0.3 mg/ Kg increments every one to two weeks to achieve a maintenance dose of 1 - 5 mg/ Kg (once a day or two divided doses) to a maximum of 200 mg/day.
Add-on therapy without Valproate	This dosage regimen should be used with: Phenytoin Carbamazepine Phenobarbitone Primidone Or with other inducers of Lamotrigine glucuronidation (see Drug interaction)	0.6 mg/ Kg (two divided doses)	1.2 mg/ Kg (two divided doses)	1.2 mg/ Kg increments every one to two weeks to achieve a maintenance dose of 5 - 15 mg/ Kg (once a day or two divided doses) to a maximum of 400 mg/day.
	This dosage regimen should be used with other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction)	0.3 mg/ Kg (one or two divided doses)	0.6 mg/ Kg (one or two divided doses)	0.6 mg/ Kg increments every one to two weeks to achieve a maintenance dose of 1 - 10 mg/ Kg (once a day or two divided doses) to a maximum of 200 mg/day.
In patients taking AEDs where the pharmacokinetic interaction with Lamotrigine is currently not known (see Drug interaction), the treatment regimen as recommended for Lamotrigine with concurrent Valproate should be used.				
*If calculated daily dose in patients taking Valproate is 1 to 2 mg, then 2 mg Lamotrigine may be taken on alternate days for the first two weeks. If the calculated daily dose in patients taking Valproate is less than 1 mg, then Lamotrigine should not be administered.				

Because of the risk of rash the initial dose and subsequent dose escalation should not be exceeded (See Warning and precaution).
It is likely that patients aged two to six years will require a maintenance dose at the higher end of the recommended range.
Children aged less than 2 years
There is insufficient information on the use of Lamotrigine in children aged less than two years.
Bipolar disorder
Adults (18 years of age and over)
Because of the risk of rash, the initial dose and subsequent dose escalation should not be exceeded (see Warning and precaution).
Lamotrigine is recommended for use in bipolar patients at risk for a future depressive episode.
The following transition regimen should be followed to prevent recurrence of depressive episodes. The transition regimen involves escalating the dose of Lamotrigine to a maintenance stabilization dose over six weeks (see Table 3) after which other psychotropic and/or anti-epileptic drugs can be withdrawn, if clinically indicated (see Table 4).
Adjunctive therapy should be considered for the prevention of manic episodes, as efficacy with Lamotrigine in mania has not been conclusively established.

Table 3: Recommended dose escalation to the maintenance total daily stabilization dose for adults (18 years of age and over) treated for Bipolar Disorder

Treatment Regimen	Weeks 1 - 2	Weeks 3 - 4	Week 5	Target Stabilization Dose (Week 6)**
a) Adjunct therapy with inhibitors of Lamotrigine glucuronidation e.g. Valproate	12.5 mg (given 25 mg alternate days)	25 mg (once a day)	50 mg (once a day or two divided doses)	100 mg (once a day or two divided doses) (maximum daily dose of 200 mg)
b) Adjunct therapy with inhibitors of Lamotrigine glucuronidation in patients NOT taking inhibitors such as Valproate This dosage regimen should be used with: Phenytoin Carbamazepine Phenobarbitone Primidone Or with other inducers of Lamotrigine glucuronidation (see Drug interaction)	50 mg (once a day)	100 mg (two divided doses)	200 mg (two divided doses)	300 mg in week 6, increasing to 400 mg/ day if necessary in week 7 (two divided doses)
c) Monotherapy with Lamotrigine OR Adjunctive therapy in patients taking other medications that do not significantly induce or inhibit Lamotrigine glucuronidation (see Drug interaction)	25 mg (once a day)	50 mg (once a day or two divided doses)	100 mg (once a day or two divided doses)	200 mg (Range 100 - 400 mg) (once a day or two divided doses)
Note: In patients taking AEDs where the pharmacokinetic interaction with Lamotrigine is currently not known, the dose escalation as recommended for Lamotrigine with concurrent Valproate should be used.				

**The target stabilization dose will alter depending on clinical purpose.

a) Adjunct therapy with inhibitors of Lamotrigine glucuronidation e.g. Valproate
In patients taking glucuronidation inhibiting concomitant drugs such as Valproate, the initial Lamotrigine dose is 25 mg every alternate day for two weeks, followed by 25 mg once a day for two weeks. The dose should be increased to 50 mg once a day (or in two divided doses) in week 5. The usual target dose to achieve optimal response is 100 mg/day given once a day or in two divided doses. However, the dose can be increased to a maximum daily dose of 200 mg, depending on clinical response.

b) Adjunct therapy with inducers of Lamotrigine glucuronidation in patients NOT taking inhibitors such as Valproate. This dosage regimen should be used with Phenytoin, Carbamazepine, Phenobarbitone, Primidone and other drugs known to induce Lamotrigine glucuronidation (see Drug interaction).
In those patients currently taking drugs that induce Lamotrigine glucuronidation and NOT taking Valproate, the initial Lamotrigine dose is 50 mg once a day for two weeks, followed by 100 mg/day given in two divided doses for two weeks. The dose should be increased to 200 mg/day given as two divided doses in week 5. The dose may be increased in week 6 to 300 mg/day however, the usual target dose to achieve optimal response is 400 mg/day given in two divided doses which may be given from week 7.

c) Monotherapy with Lamotrigine OR adjunctive therapy in patients taking other medications that do not significantly induce or inhibit Lamotrigine glucuronidation (see Drug interaction).
The initial Lamotrigine dose is 25 mg once a day for two weeks, followed by 50 mg once a day (or in two divided doses) for two weeks. The dose should be increased to 100 mg/day in week 5. The usual target dose to achieve optimal response is 200 mg/day given once a day or as two divided doses. However, a range of 100 to 400 mg was used in clinical trials.
Once the target daily maintenance stabilization dose has been achieved, other psychotropic medications may be withdrawn as laid out in the dosage schedule below (see Table 4).

Table 4: Maintenance stabilization total daily dose in adults (18 years of age and over) with BIPOLAR DISORDER following withdrawal of concomitant psychotropic or anti- epileptic drugs

Treatment Regimen	Week 1	Week 2	Week 3 onwards*
a) Following withdrawal of inhibitors of Lamotrigine glucuronidation e.g. Valproate	Double the stabilization dose, not exceeding 100 mg/week i.e. 100 mg/day target stabilization dose will be increased in week 1 to 200 mg/day	Maintain this dose (200 mg/day) (two divided doses)	
b) Following withdrawal of inducers of Lamotrigine glucuronidation depending on original dose This dose regimen should be used with:	400 mg	300 mg	200 mg
	300 mg	225 mg	150 mg
	200 mg	150 mg	100 mg

PM Specification	Product	Code No.	Dimension	Packaging Type	Thickness
	LAMOGA (25 mg, 50 mg, 100 mg)	ILMY 0091	W 29 x L 17 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm.)
	Designed by: (PDM, PDC-M, PDC-A, PDO-M, PDO-A)	Checked by: (PDM)	Approved by: (MD (Only Pattern and Colors))		

Approved by: PLC: (Date) LRA: (Date) IRA: (Date) QCC-PM: (Date) CUSTOMER/SALES DEPT.: (Date)

Print : Fit to Page

LAMOGA 50_Insert_Front_Back_ILMY 0091 _BY...Ann 03/10/2025

Phenytoin Carbamazepine Phenobarbitone Primidone Or with other inducers of Lamotrigine glucuronidation (see Drug interaction)			
c) Following withdrawal of other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction)	Maintain target dose achieved in dose escalation (200 mg/day) (two divided doses) (Range 100-400 mg)		
Note: In patients taking AEDs where the pharmacokinetic interaction with Lamotrigine is currently not known, the treatment regimen recommended for Lamotrigine is to initially maintain the current dose and adjust the Lamotrigine treatment based on clinical response.			

*Dose may be increased to 400 mg/day as needed.

a) Following withdrawal of adjunct therapy with inhibitors of Lamotrigine glucuronidation e.g. Valproate
The dose of Lamotrigine should be increased to double the original target stabilization dose and maintained at this, once Valproate has been terminated.

b) Following withdrawal of adjunct therapy with inducers of Lamotrigine glucuronidation depending on original maintenance dose. This regimen should be used with Phenytoin, Carbamazepine, Phenobarbitone, Primidone or other drugs known to induce Lamotrigine glucuronidation (see Drug interaction).
The dose of Lamotrigine should be gradually reduced over three weeks as the glucuronidation inducer is withdrawn.

c) Following withdrawal of adjunct therapy with other medications that do not significantly inhibit or induce Lamotrigine glucuronidation (see Drug interaction).
The target dose achieved in the dose escalation program should be maintained throughout withdrawal of the other medication.

Adjustment of Lamotrigine daily dosing in patients with BIPOLAR DISORDER following addition of other medications
There is no clinical experience in adjusting the Lamotrigine daily dose following the addition of other medications. However, based on drug interaction studies, the following recommendations can be made (see Table 5 below):

Table 5: Adjustment of Lamotrigine daily dosing in adults (18 years of age and over) with BIPOLAR DISORDER following the addition of other medications

Treatment Regimen	Current Lamotrigine Stabilization Dose (mg/day)	Week 1	Week 2	Week 3 onwards
a) Addition of inhibitors of Lamotrigine glucuronidation e.g. Valproate, depending on original dose of Lamotrigine	200 mg	100 mg	Maintain this dose (100 mg/day)	
	300 mg	150 mg	Maintain this dose (150 mg/day)	
	400 mg	200 mg	Maintain this dose (200 mg/day)	
b) Addition of inducers of Lamotrigine glucuronidation in patients NOT taking Valproate and depending on original dose of Lamotrigine.	200 mg	200 mg	300 mg	400 mg
	150 mg	150 mg	225 mg	300 mg
	100 mg	100 mg	150 mg	200 mg
This dosage regimen should be used with: Phenytoin Carbamazepine Phenobarbitone Primidone Or with other inducers of Lamotrigine glucuronidation (see Drug interaction)				
Note: In patients taking AEDs where the pharmacokinetic interaction with Lamotrigine is currently not known, the treatment regimen as recommended for Lamotrigine with concurrent Valproate should be used.				

Discontinuation of Lamotrigine in adult patients With Bipolar Disorder
Patients may terminate Lamotrigine without a step-wise reduction of dose.
Children and adolescents (less than 18 years of age)
Lamotrigine is not indicated for use in bipolar disorder in children and adolescents aged less than 18 years (see Warning and precaution). Safety and efficacy of Lamotrigine in bipolar disorder has not been established in this age group. Therefore, a dosage recommendation cannot be made.
General Dosing Recommendations for Lamotrigine in Special Patient Populations
Women taking hormonal contraceptives
a) Starting Lamotrigine in patients already taking hormonal contraceptives:
Although an oral contraceptive has been shown to increase the clearance of Lamotrigine (see Warning and precaution and Drug interaction), no adjustments to the recommended dose escalation guidelines for Lamotrigine should be necessary solely based on the use of hormonal contraceptives. Dose escalation should follow the recommended guidelines based on whether Lamotrigine is added to Valproate (an inhibitor of Lamotrigine glucuronidation), or to an inducer of Lamotrigine glucuronidation, or whether Lamotrigine is added in the absence of Valproate or an inducer of Lamotrigine glucuronidation (see Table 1 for epilepsy and Table 3 for bipolar disorder patients).

b) Starting hormonal contraceptives in patients already taking maintenance doses of Lamotrigine and NOT taking inducers of Lamotrigine glucuronidation:
The maintenance dose of Lamotrigine will in most cases need to be increased by as much as two-fold (see Warning and precaution and Drug interaction). It is recommended that from the time that the hormonal contraceptive is started, the Lamotrigine dose is increased by 50 to 100 mg/day every week, according to the individual clinical response. Dose increases should not exceed this rate, unless the clinical response supports larger increases.

c) Stopping hormonal contraceptives in patients already taking maintenance doses of Lamotrigine and NOT taking inducers of Lamotrigine glucuronidation:
The maintenance dose of Lamotrigine will in most cases need to be decreased by as much as 50% (see Warning and precaution and Drug interaction). It is recommended to gradually decrease the daily dose of Lamotrigine by 50 to 100 mg each week (at a rate not exceeding 25% of the total daily dose per week) over a period of 3 weeks, unless the clinical response indicates otherwise.
Use with Atazanavir/ Ritonavir
Although Atazanavir/Ritonavir has been shown to reduce Lamotrigine plasma concentrations (see Drug interaction), no adjustments to the recommended dose escalation guidelines for Lamotrigine should be necessary solely based on the use of Atazanavir/Ritonavir. Dose escalation should follow the recommended guidelines based on whether Lamotrigine is added to Valproate (an inhibitor of Lamotrigine glucuronidation), or to an inducer of Lamotrigine glucuronidation, or whether Lamotrigine is added in the absence of Valproate or an inducer of Lamotrigine glucuronidation.
In patients already taking maintenance doses of Lamotrigine and not taking glucuronidation inducers, the Lamotrigine dose may need to be increased if Atazanavir/Ritonavir is added, or decreased if Atazanavir/Ritonavir is discontinued.
Elderly (over 65 years of age)
No dosage adjustment from recommended schedule is required. The pharmacokinetics of Lamotrigine in this age group do not differ significantly from a non-elderly adult population.
Hepatic impairment
Initial, escalation and maintenance doses should generally be reduced by approximately 50% in patients with moderate (Child-Pugh grade B) and 75% in severe (Child-Pugh grade C) hepatic impairment. Escalation and maintenance doses should be adjusted according to clinical response (see Pharmacokinetics).
Renal impairment
Caution should be exercised when administering Lamotrigine to patients with renal failure. For patients with end-stage renal failure, initial doses of Lamotrigine should be based on patients' AED regimen; reduced maintenance doses may be effective for patients with significant renal functional impairment (see Warning and precaution). For more detailed pharmacokinetic information (see Pharmacokinetics).

✧ WARNING AND PRECAUTION:

Skin rash
The risk of serious skin rashes in children is higher than in adults.
In children, the initial presentation of a rash can be mistaken for an infection; physicians should consider the possibility of a reaction to Lamotrigine treatment in children that develop symptoms of rash and fever during the first eight weeks of therapy.
Additionally the overall risk of rash appears to be strongly associated with:
- High initial doses of Lamotrigine and exceeding the recommended dose escalation of Lamotrigine therapy
- Concomitant use of Valproate

Caution is also required when treating patients with a history of allergy or rash to other AEDs as the frequency of non-serious rash after treatment with Lamotrigine was approximately three times higher in these patients than in those without such history.
All patients (adults and children) who develop a rash should be promptly evaluated and Lamotrigine withdrawn immediately unless the rash is clearly not related to Lamotrigine treatment. It is recommended that Lamotrigine tablets not be restarted in patients who have discontinued due to rash associated with prior treatment with Lamotrigine unless the potential benefit clearly outweighs the risk. If the patient has developed SJS, TEN or DRESS with the use of Lamotrigine, treatment with Lamotrigine must not be restarted in this patient at any time.
Rash has also been reported as part of a hypersensitivity syndrome associated with a variable pattern of systemic symptoms including fever, lymphadenopathy, facial edema, abnormalities of the blood and liver and aseptic meningitis. The syndrome shows a wide spectrum of clinical severity and may, rarely, lead to disseminated intravascular coagulation and multiorgan failure. It is important to note that early manifestations of hypersensitivity (for example fever, lymphadenopathy) may be present even though rash is not evident. If such signs and symptoms are present the patient should be evaluated immediately and Lamotrigine discontinued if an alternative etiology cannot be established.
Aseptic meningitis was reversible on withdrawal of the drug in most cases, but recurred in a number of cases on re-exposure to Lamotrigine. Re-exposure resulted in a rapid return of symptoms that were frequently more severe. Lamotrigine should not be restarted in patients who have discontinued due to aseptic meningitis associated with prior treatment of Lamotrigine.
Clinical worsening and suicide risk
Suicidal ideation and behavior have been reported in patients treated with AEDs in several indications.
Potential for an increase risk of suicidal thoughts or behaviors.
Therefore patients should be monitored for signs of suicidal ideation and behaviors and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behavior emerge.
In patients with bipolar disorder, worsening of depressive symptoms and/or the emergence of suicidality may occur whether or not they are taking medications for bipolar disorder, including Lamotrigine tablets. Therefore patients receiving Lamotrigine tablets for bipolar disorder should be closely monitored for clinical worsening (including development of new symptoms) and suicidality, especially at the beginning of a course of treatment, or at the time of dose changes. Certain patients, such as those with a history of suicidal behavior or thoughts, young adults, and those patients exhibiting a significant degree of suicidal ideation prior to commencement of treatment, may be at a greater risk of suicidal thoughts or suicide attempts, and should receive careful monitoring during treatment.
Consideration should be given to changing the therapeutic regimen, including possibly discontinuing the medication, in patients who experience clinical worsening (including development of new symptoms) and/or the emergence of suicidal ideation/behavior, especially if these symptoms are severe, abrupt in onset, or were not part of the patient's presenting symptoms.
Hormonal contraceptives
Effects of hormonal contraceptives on Lamotrigine efficacy
The use of an Ethinylestradiol/Levonorgestrel (30 μg/150 μg) combination increases the clearance of Lamotrigine by approximately two-fold resulting in decreased Lamotrigine levels. A decrease in Lamotrigine levels has been associated with loss of seizure control. Following titration, higher maintenance doses of Lamotrigine (by as much as two-fold) will be needed in most cases to attain a maximal therapeutic response. When stopping hormonal contraceptives, the clearance of Lamotrigine may be halved. Increases in Lamotrigine concentrations may be associated with dose-related adverse events. Patients should be monitored with respect to this (see Dosage and administration).
Effects of Lamotrigine on hormonal contraceptive efficacy
The possibility of these changes (serum FSH and LH) resulting in decreased contraceptive efficacy in some patients taking hormonal preparations with Lamotrigine cannot be excluded. Therefore patients should be instructed to promptly report changes in their menstrual pattern, i.e. breakthrough bleeding.
Dihydrofolate reductase
Lamotrigine has a slight inhibitory effect on Dihydrofolic acid reductase, hence there is a possibility of interference with folate metabolism during long-term therapy. However, during prolonged human dosing, Lamotrigine did not induce significant changes in the hemoglobin concentration, mean corpuscular volume, or serum or red blood cell Folate concentrations up to 1 year or red blood cell Folate concentrations for up to 5 years.
Renal failure
Accumulation of the glucuronide metabolite is to be expected; caution should therefore be exercised in treating patients with renal failure.

Patients taking other preparations containing Lamotrigine
Lamotrigine should not be administered to patients currently being treated with any other preparation containing Lamotrigine without consulting a doctor.
Precautions relating to epilepsy
As with other AEDs, abrupt withdrawal of Lamotrigine may provoke rebound seizures. Unless safety concerns (for example rash) require an abrupt withdrawal, the dose of Lamotrigine should be gradually decreased over a period of two weeks.
A clinically significant worsening of seizure frequency instead of an improvement may be observed. In patients with more than one seizure type, the observed benefit of control for one seizure type should be weighed against any observed worsening in another seizure type.
Myoclonic seizures may be worsened by Lamotrigine.
Precautions relating to bipolar disorder
Children and adolescents below 18 years
Treatment with antidepressants is associated with an increased risk of suicidal thinking and behavior in children and adolescents with major depressive disorder and other psychiatric disorders.
Warning and precaution[2]:
Hemaphagocytic lymphohistiocytosis (HLH) has occurred in patients taking Lamotrigine (see Side effect). HLH is a syndrome is a pathological immune activation, which can be life threatening, characterized by clinical signs and symptoms such as fever, rash, neurological symptoms, hepatosplenomegaly, lymphadenopathy, cytopenias, high serum ferritin, hypertriglyceridemia and abnormalities of liver function and coagulation. Symptoms occur generally within 4 weeks of treatment initiation. Immediately evaluate patients who develop these signs and symptoms and consider a diagnosis of HLH. Lamotrigine should be discontinued unless an alternative etiology can be established.
Brugada-type ECG
A very rare association with Brugada-type ECG may be observed, although a casual relationship has not been established. Therefore careful consideration should be given before using Lamotrigine in patients with Brugada syndrome.

✧ PREGNANCY AND LACTATION[3]:

Risk related to antiepileptic drugs in general
Specialist advice should be given to women who are of childbearing potential. The antiepileptic treatment should be reviewed when a woman is planning to become pregnant. In women being treated for epilepsy, sudden discontinuation of AED therapy should be avoided as this may lead to breakthrough seizures that could have serious consequences for the woman and the unborn child.
Monotherapy should be preferred whenever possible because therapy with multiple AEDs could be associated with a higher risk of congenital malformations than monotherapy, depending on the associated antiepileptics.
Risk related to Lamotrigine
Pregnancy
If therapy with Lamotrigine is considered necessary during pregnancy, the lowest possible therapeutic dose is recommended.
Lamotrigine has a slight inhibitory effect on Dihydrofolic acid reductase and could therefore theoretically lead to an increased risk of embryofetal damage by reducing folic acid levels. Intake of folic acid when planning pregnancy and during early pregnancy may be considered.
Physiological changes during pregnancy may affect Lamotrigine levels and/or therapeutic effect.
Lactation
The potential benefits of breastfeeding should be weighed against the potential risk of adverse effects occurring in the infant. Should a woman decide to breastfeed while on therapy with Lamotrigine, the infant should be monitored for adverse effects.

✧ SIDE EFFECT[1]:

Following are the adverse reactions identified from Epilepsy and Bipolar disorders:
Epilepsy
Skin and subcutaneous tissue disorders
Very common : Skin rash
Rare : Stevens Johnson syndrome , erythema multiforme
Very rare : Toxic epidermal necrolysis
The overall risk of rash appears to be strongly associated with:
- High doses of Lamotrigine and exceeding the recommended dose escalation of Lamotrigine therapy (See Dosage and administration)
- Concomitant use of Valproate (see Dosage and administration)

Blood and lymphatic system disorders
Very rare : Hematological abnormalities (including, neutropenia, leucopenia, anemia, thrombocytopenia, pancytopenia, aplastic anemia, agranulocytosis), lymphadenopathy
Hematological abnormalities and lymphadenopathy may or may not be associated with DRESS/ Hypersensitivity Syndrome (see Warnings and precautions and Immune System Disorders**).
Immune system disorders
Very rare : DRESS / Hypersensitivity syndrome** including such symptoms as, fever, lymphadenopathy, facial edema, abnormalities of the blood, liver and kidney
**Rash has also been reported as part of this syndrome which shows a wide spectrum of clinical severity and may, rarely, lead to disseminated intravascular coagulation (DIC) and multi-organ failure. It is important to note that early manifestations of hypersensitivity (e.g. fever, lymphadenopathy) may be present even though rash is not evident. If such signs and symptoms are present the patient should be evaluated immediately, and Lamotrigine discontinued if an alternative etiology cannot be established.
Psychiatric disorders
Common : Aggression, irritability
Very rare : Tics, hallucinations, confusion
Nervous system disorders
Very common : Headache
Common : Somnolence, insomnia, dizziness, tremor
Uncommon : Ataxia
Rare : Nystagmus
Eye disorders
Uncommon : Diplopia, blurred vision
Gastrointestinal disorders
Common : Nausea, vomiting, diarrhea
Hepatobiliary disorders
Very rare : Increased liver function tests, hepatic dysfunction, hepatic failure
Hepatic dysfunction usually occurs in association with hypersensitivity reactions but isolated cases have been reported without overt signs of hypersensitivity.
Musculoskeletal and connective tissue disorders
Very rare : Lupus-like reactions
General disorders and administration site conditions
Very rare : Tiredness

Bipolar Disorder
Skin and subcutaneous tissue disorders
Very common : Skin rash
Rare : Stevens Johnson syndrome , erythema multiforme
Nervous system disorders
Very common : Headache
Common : Agitation, somnolence, dizziness
Musculoskeletal and connective tissue disorders
Common : Arthralgia
General disorders and administration site conditions
Common : Pain, back pain
Post-marketing
Blood and Lymphatic system disorder
Very rare : Hemaphagocytic lymphohistiocytosis (see Warnings and Precautions)

✧ EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

As there is individual variation in response to all AED therapy, patients taking Lamotrigine to treat epilepsy should consult their physician on the specific issues of driving and epilepsy. Based on the side effects sited, patients should see how Lamotrigine therapy affects them before driving or operating machinery.

✧ CONTRAINDICATION:

Hypersensitivity to Lamotrigine or to any of the excipients.

✧ DRUG INTERACTION:

Interaction studies have only been performed in adults.
UDP-glucuronyl transferases have been identified as the enzymes responsible for metabolism of Lamotrigine. There is no evidence that Lamotrigine causes clinically significant induction or inhibition of hepatic oxidative drug-metabolizing enzymes, and interactions between Lamotrigine and medicinal products metabolized by cytochrome P450 enzymes are unlikely to occur. Lamotrigine may induce its own metabolism but the effect is modest and unlikely to have significant clinical consequences.

Table 6: Effects of other medicinal products on glucuronidation of Lamotrigine

Medicinal products that significantly inhibit glucuronidation of Lamotrigine	Medicinal products that significantly induce glucuronidation of Lamotrigine	Medicinal products that do not significantly inhibit or induce glucuronidation of Lamotrigine
Valproate	Phenytoin	Oxcarbazepine
-	Carbamazepine	Felbamate
-	Phenobarbitone	Gabapentin
-	Primidone	Levetiracetam
-	Rifampicin	Pregabalin
-	Lopinavir/Ritonavir	Topiramate
-	Ethinyloestradiol/ Levonorgestrel combination**	Zonisamide
-	Atazanavir/Ritonavir*	Lithium
-	-	Buproprion
-	-	Olanzapine
-	-	Aripiprazole

*For dosing guidance (see Dosage and administration)
**Other oral contraceptive and HRT treatments have not been studied, though they may similarly affect Lamotrigine pharmacokinetic parameters (see Dosage and administration and Warning and precaution).

✧ OVERDOSE AND TREATMENT:

Symptoms and signs
Acute ingestion of doses in excess of 10 to 20 times the maximum therapeutic dose has been reported, including fatal cases. Overdose has resulted in symptoms including nystagmus, ataxia, impaired consciousness, grand mal convulsion and coma. QRS broadening (intraventricular conduction delay) has also been observed in overdose patients. Broadening of QRS duration to more than 100 msec may be associated with more severe toxicity.
Treatment
In the event of overdose, the patient should be admitted to hospital and given appropriate supportive therapy. Therapy aimed at decreasing absorption (Activated Charcoal) should be performed if indicated. Further management should be as clinically indicated. There is no experience with hemodialysis as treatment of overdose.

✧ STORAGE:

Store at temperature of not more than 30°C.

✧ DOSAGE FORM AND PACKAGING AVAILABLE:

25 mg Tablet, Blister 3x10's
50 mg Tablet, Blister 3x10's
100 mg Tablet, Blister 3x10's

✧ DATE OF REVISION:

September 02, 2025

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

C-MY/020925-06 (AR)
(www.emc-medicines.org.uk; [1]DCA reg-Lamictal; [2]Lamotrigine SLR; [3]DCA 2nd cor , 8/15)
ILMY 0091